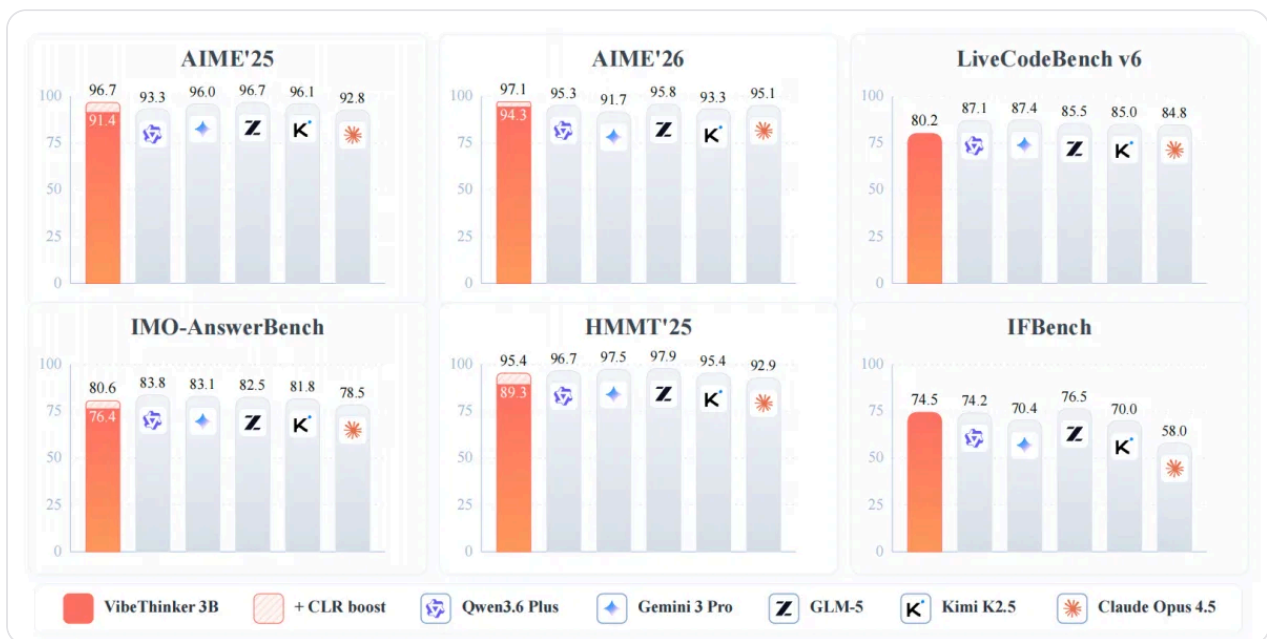


VibeThinker-3B: The 3B Model That Goes Toe-to-Toe with Trillion-Parameter AI



Links & Resources

- [Hugging Face \(3B\)](#)
- [Hugging Face \(1.5B\)](#)
- [GitHub](#)
- [ModelScope](#)
- [Technical Report \(3B\)](#)
- [Technical Report \(1.5B\)](#)

A 3-billion-parameter model just scored **94.3 on AIME 2026** and passed **123 out of 128 LeetCode problems** on the first try. It's called **VibeThinker-3B**, it's from Weibo (China's Twitter), and it's free on Hugging Face under an MIT license.

For context, models that hit these numbers typically have 300 billion to over a trillion parameters. VibeThinker has 3 billion. That's not a typo - it's 100 to 300 times smaller.

The Numbers That Shouldn't Be Possible

VibeThinker-3B reaches **76.4 on IMO-AnswerBench** - a benchmark of 400 IMO-level math problems - with just 3B parameters. With their test-time scaling strategy (CLR), it climbs to **80.6**.

For comparison, here are models hundreds of times larger:

MODEL	PARAMETERS	IMO-ANSWERBENCH
VibeThinker-3B + CLR	3B	80.6
DeepSeek V3.2	671B	78.3
Kimi K2.5	1T+	81.8
GLM-5	744B	82.5

A 3B model trading blows with trillion-parameter systems. That's the headline.

Full Benchmark Results

Math

BENCHMARK	VIBETHINKER-3B	WITH CLR
AIME 2026	94.3	97.1
HMMT 2025	89.3	95.4

BENCHMARK	VIBETHINKER-3B	WITH CLR
BruMO 2025	-	99.2
IMO-AnswerBench	76.4	80.6

Coding

BENCHMARK	VIBETHINKER-3B
LiveCodeBench v6	80.2 (Pass@1)
LeetCode Contests (Apr 25 - May 31, 2026)	123/128 passed (96.1% acceptance)

That LeetCode number is especially telling because it's out-of-distribution - these are recent unseen contests the model was never trained on. It's not memorizing solutions. It's reasoning its way through problems it's never seen.

How It Competes with Frontier Models

With CLR (Claim-Level Reliability Assessment) enabled - a test-time scaling strategy that lets the model verify its own reasoning - VibeThinker-3B enters the performance range of top-tier systems like Qwen3.6 Plus, Gemini 3 Pro, GLM-5, and Kimi K2.5 on verifiable reasoning benchmarks.

It doesn't match them on everything. Open-domain knowledge, general conversation, and long-tail understanding still favor larger models. But on math, coding, and STEM reasoning - tasks where the answer can be verified - this 3B model plays in the same league.

Why It Works - The Parametric Compression Hypothesis

The team behind VibeThinker proposes a compelling theory: **different capabilities scale with parameters in fundamentally different ways.**

Verifiable reasoning - multi-step math, constraint satisfaction, self-correction, answer verification - is a highly compressible, parameter-dense capability. When the task space is structured and feedback signals are reliable, compact models can carry near-frontier reasoning ability.

Open-domain knowledge, on the other hand, needs massive parameter counts to broadly cover facts, concepts, and world knowledge. You can't compress "know everything about everything" the way you can compress "reason step-by-step."

This is why VibeThinker crushes math olympiads and LeetCode but isn't trying to replace ChatGPT for general conversation. The model does one thing and does it at frontier level.

The Training Pipeline

VibeThinker-3B is built on Qwen2.5-Coder-3B and trained using the **Spectrum-to-Signal Principle (SSP)** - a four-stage post-training pipeline:

Stage 1 - Curriculum SFT (Supervised Fine-Tuning) Two phases: broad capability coverage first (math, code, STEM, dialogue), then harder, longer-horizon reasoning samples. Uses Diversity-Exploring Distillation to preserve multiple valid solution paths instead of collapsing to one.

Stage 2 - Multi-Domain Reasoning RL Reinforcement learning applied sequentially to math, code, and STEM reasoning tasks using MaxEnt-Guided Policy Optimization (MGPO). Training runs in a single 64K context window to preserve complete reasoning chains.

Stage 3 - Offline Self-Distillation High-quality reasoning traces from the RL checkpoints are filtered and distilled back into a unified model. A learning-potential score prioritizes traces that are correct but not yet well-modeled - teaching the student what it still gets wrong.

Stage 4 - Instruct RL Final stage improves controllability on user-facing prompts using rule-based validators and rubric-based reward models.

The Cost

For reference, the 1.5B predecessor cost just **\$7,800** to post-train - compared to \$294K for DeepSeek R1 and \$535K for MiniMax-M1. A 30-60x reduction in training cost for competitive performance.

How to Run It

Option 1: vLLM (Recommended)

```
pip install vllm
vllm serve "WeiboAI/VibeThinker-3B"
```

Option 2: SGLang

```
pip install sglang
python3 -m sglang.launch_server \
  --model-path "WeiboAI/VibeThinker-3B" \
  --host 0.0.0.0 --port 30000
```

Option 3: Docker

```
docker model run hf.co/WeiboAI/VibeThinker-3B
```

Recommended Settings

```
temperature: 1.0
top_p: 0.95
top_k: -1 (disabled)
max_new_tokens: 40960 (or 102400 for 3B)
```

Requires `transformers>=4.54.0`. For best inference performance, use vLLM 0.10.1 or SGLang `>=0.4.9.post6`.

The VibeThinker Family

MODEL	PARAMETERS	BASE	HIGHLIGHTS
VibeThinker-3B	3B	Qwen2.5-Coder-3B	Frontier-level math/code, 96.1% LeetCode acceptance
VibeThinker-1.5B	1.5B	Qwen2.5-Coder-1.5B	Beats DeepSeek R1 (671B) on AIME24, AIME25, HMMT25. Trained for \$7,800

Both models are MIT licensed - fully open, commercial use allowed.

What This Means

VibeThinker-3B isn't trying to replace GPT-5.5 or Claude. It's proving something more interesting: that for specific, verifiable tasks - the kind of math and coding problems people pay \$20/month to solve - a model small enough to run on a laptop can match systems that cost millions to train and run.

The coding power that lives behind a \$20/month subscription? It's now a free download on Hugging Face.

